

Radial Single-Kernel f_i Optimization Report

19 March 2026

This report summarizes the radial single-kernel notebook and the later constrained version that was added to make the kernel shape more meaningful.

Results from the Notebook

Original radial fit (`abs_max`, outer radii frozen to seed):

Val: AUC 0.8898, accuracy at 0.5 = 0.8182.

Test: AUC 0.8825, accuracy at 0.5 = 0.8000

Learned radial kernel:

Val: AUC 0.9189, accuracy at 0.5 = 0.8545

Test: AUC 0.9286, accuracy at 0.5 = 0.8668

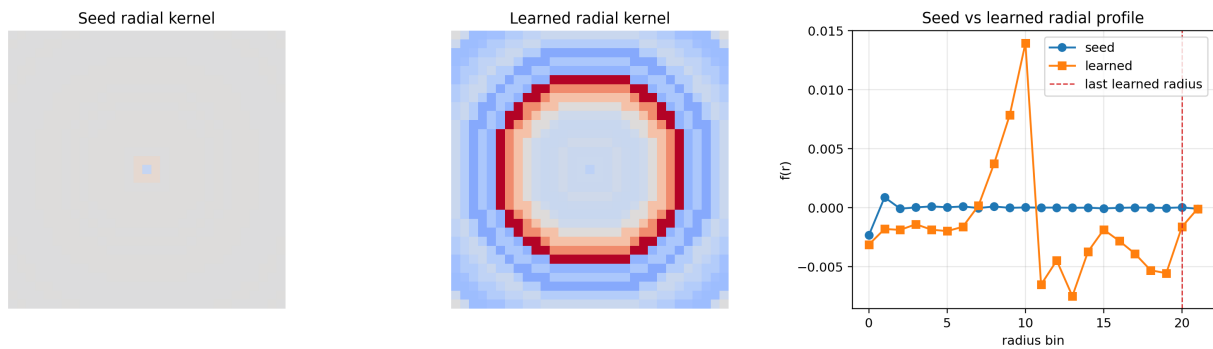


Figure 1: Original learned radial kernel from the notebook with f_i limit = 20.

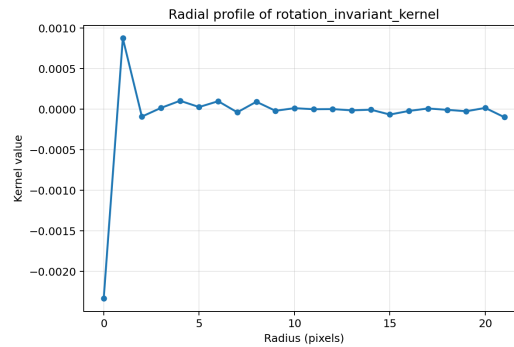


Figure 2: Radial profile of the original rotation-invariant kernel from 21st Feb report

The learned kernel has a similar starting shape (focused on 0 - 5) where it is increasing then decreasing and then plateaued.

Clarifications

Why does it look different from the earlier preset-family kernel?

The earlier method was restricted by preset kernel families. This approach searches in a different space and learns completely free radial profile directly from data, so a different shape is expected.

Why is the learned profile not always decreasing with radius?

The original loss does not force the profile to decrease with radius. It only rewards what improves classification, and outer rings cover more pixels, so they can still receive large weight.

Why can different runs give different kernels?

The kernel is not uniquely determined by the classification loss. Several different radial profiles can give similar responses and similar accuracy, and this ambiguity becomes stronger when `abs_max` removes sign information. (Added constraints to get a converging kernel in the next section)

Changes to make the kernel converging

The constrained version added four changes:

- Replace `abs_max` with a signed response.
- Zero out $r > R_{\max}$ instead of freezing those radii to the seed.
- Add monotone and decay regularization.
- Normalize kernel scale and fix the sign convention.

Signed Compact Kernel Results

Signed compact fit (`signed_peak`, hard cutoff at $R_{\max} = 20$):

Val: AUC 0.8878, accuracy at 0.5 = 0.8150

Test: AUC 0.8808, accuracy at 0.5 = 0.8011

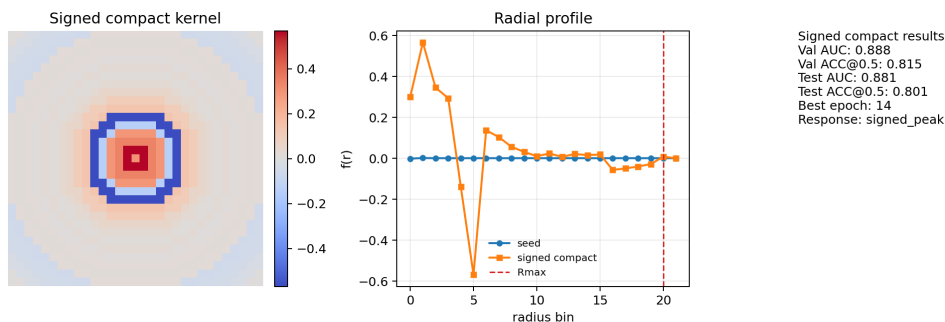


Figure 3: Final signed compact kernel and its radial profile. This is the constrained kernel used for the later exact heatmap.

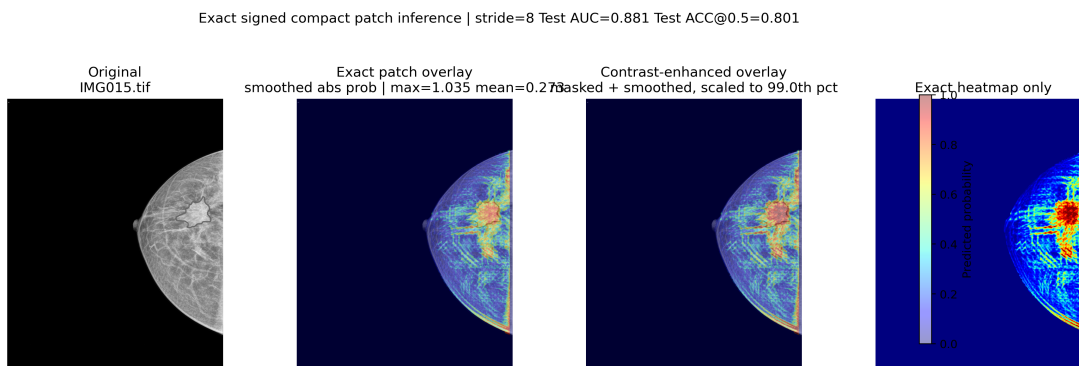


Figure 4: Heatmap of the signed compact kernel

Summary

The original `abs_max` radial fit gives the best classification numbers in this notebook, with test AUC **0.9286** and test accuracy **0.8668**. However, that setup does not identify a unique kernel shape very well. The later signed compact version gives slightly lower classification performance, with test AUC **0.8808** and test accuracy **0.8011**, but it makes the kernel itself easier to interpret because the response is signed, the support is explicitly limited, and the profile is regularized.